MINISTRY OF INNOVATION AND TECHNOLOGY

MinT Training Management System

Online Training Management System for the Ministry of Innovation and Technology (MinT), Ethiopia

Project Type	Internship / Software Development Project
Organization	Ministry of Innovation and Technology (MinT), Ethiopia
System Type	Web-based Training Management System
Frontend	React.js
Backend	Node.js + Express.js
Database	PostgreSQL
Authentication	JWT + bcrypt

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Contents

1. Introduction
2. Background of the System
3. Problem Statement
4. Purpose of the System
5. Objectives
5.1 General Objective
5.2 Specific Objectives
6. Scope of the System
6.1 In-Scope
6.2 Out-of-Scope
7. Target Users
7.1 Administrator
7.2 Trainer
7.3 Trainee
8. Product Perspective
9. System Architecture
10. System Components
10.1 Frontend
10.2 Backend API
10.3 PostgreSQL Database
11. Functional Requirements
12. Training Assignment and Enrollment Workflow
13. Learning Content and Progress
14. Attendance and Certificates
15. Reports and Dashboard Requirements
16. Non-Functional Requirements
17. Security Requirements
18. Error Handling
19. Testing Requirements
20. Future Enhancements
21. Conclusion

1. Introduction

The MinT Training Management System is a web-based application designed to support the planning, administration, delivery, monitoring, and completion of short-term training programs conducted by the Ministry of Innovation and Technology (MinT), Ethiopia. The system provides separate role-based experiences for administrators, trainers, and trainees.

The application replaces fragmented or manual training-management activities with a centralized platform. Administrators can manage users, trainers, training programs, department assignments, enrollment requests, attendance, certificates, and reports. Trainers can manage their assigned training content and participants, while trainees can register, view eligible programs, request enrollment, study published learning materials, track progress, and obtain certificates when the defined completion requirements are satisfied.

The system follows a professional MinT government-style interface and uses a React frontend, an Express/Node.js backend API, and PostgreSQL for persistent data storage. Authentication and authorization are implemented with JWT-based sessions and secure password hashing.

2. Background of the System

Short-term training programs may involve many trainees, trainers, departments, learning materials, attendance records, and completion documents. When these activities are handled manually or through disconnected tools, it becomes difficult to maintain consistent records, monitor requests, verify participation, and provide timely reports.

The proposed system centralizes these activities in one application. It also introduces department-aware training assignment so that a training can be made available to one department, multiple departments, or all departments without incorrectly rejecting eligible trainees. Enrollment requests are managed through a controlled Pending, Approved, Rejected, or Cancelled workflow.

3. Problem Statement

• Training information, enrollment activities, and participant records can be difficult to manage consistently without a centralized system.

• Trainees need a clear way to discover training programs that are relevant to their department.

• Training assignments may apply to one, several, or all departments and therefore require explicit eligibility logic.

• Enrollment requests require a controlled approval process so that trainees cannot approve their own requests.

• Administrators need visibility into trainee requests, statuses, request dates, departments, and related actions.

• Trainers need a controlled way to publish learning materials for their assigned programs.

• Training progress, attendance, and certificate generation require reliable database-backed records.

• The public website and role-based dashboards must remain visually consistent, responsive, and secure.

4. Purpose of the System

The primary purpose of the MinT Training Management System is to provide an integrated, secure, and user-friendly platform for managing short-term training programs from registration and

enrollment through learning, attendance, progress monitoring, and certification.

5. Objectives

5.1 General Objective

To develop a web-based training management system that centralizes training administration, department-based training assignment, enrollment approval, learning delivery, progress tracking, attendance, certification, and reporting.

5.2 Specific Objectives

• Provide secure authentication and role-based authorization for administrators, trainers, and trainees.

• Allow administrators to create and manage training programs and trainer accounts.

• Allow trainees to register and identify their department.

• Support training assignments to a single department, multiple departments, or all departments.

• Allow eligible trainees to submit training enrollment requests.

• Provide a Pending → Approved/Rejected workflow controlled by administrators.

• Allow trainees to cancel their own Pending requests while preventing cancellation through that function after approval.

• Allow administrators to monitor trainee enrollment activity and request history.

• Allow trainers and authorized administrators to publish learning modules and lessons.

• Track trainee learning progress and support completion-based certificate generation.

• Maintain attendance, participant, certificate, and reporting records.

• Provide a responsive public website with English and Amharic user-interface support.

6. Scope of the System

6.1 In-Scope

• Public home, About, Contact, and Training Programs pages.

• User registration and login.

• JWT authentication and role-based authorization.

• Administrator dashboard and management functions.

• Trainer creation and trainer management.

• Trainee management and department information.

• Training program creation, editing, viewing, and assignment.

• Department-based training eligibility.

• Enrollment/training-request submission and status management.

• Administrator approval and rejection of pending requests.

• Trainee cancellation/removal of Pending requests.

• Learning modules, lessons, and published training materials.

• Learning progress tracking and completion requirements.

• Attendance management.

• Certificate generation and certificate records.

• Reports and dashboard analytics supported by the database.

• English and Amharic interface support.

• Responsive and professional MinT-style user interface.

6.2 Out-of-Scope

• Replacing the Ministry's broader human-resources or enterprise information systems.

• Processing or exposing passwords through administrative reports.

• Unauthorized access to protected administrative functions.

• Bypassing authentication or role-based authorization.

• Unnecessary dependence on external services where an existing application mechanism is sufficient.

• Treating a placeholder contact form as a real message-delivery service when no backend destination is configured.

7. Target Users

7.1 Administrator

• Manage trainers and trainees.

• Create and manage training programs.

• Assign trainings to departments.

• View all trainee enrollment requests.

• Approve or reject pending requests.

- Monitor request status and relevant action information.
- Manage attendance, certificates, reports, and other authorized system functions.

7.2 Trainer

- Access assigned training programs.
- Manage learning modules and lessons for permitted trainings.
- Publish learning materials.
- Manage participants and attendance according to existing permissions.

7.3 Trainee

- Register and select a department.
- View training programs eligible for the trainee's department.
- Submit enrollment requests.
- View personal request status.
- Cancel personal Pending requests.
- Access approved training and published learning materials.
- Track progress and obtain certificates when requirements are met.

8. Product Perspective

The system is a full-stack web application consisting of a browser-based React client, a Node.js/Express API server, and a PostgreSQL database. The frontend communicates with protected backend endpoints using HTTP requests and an authenticated JWT. The backend applies business rules and authorization before reading or changing database records.

The public website uses a public layout with its normal header and footer. Authenticated users use role-specific dashboard layouts. This separation prevents the public Training Programs page from accidentally rendering an administrator, trainer, or trainee dashboard sidebar.

9. System Architecture

High-Level Architecture

Layer	Technology / Responsibility
Presentation Layer	React.js, React Router DOM, Tailwind CSS, responsive MinT-style UI
API Layer	Node.js + Express.js; routes, controllers, middleware, validation, business logic
Authentication	JWT authentication, bcrypt password hashing, role-based authorization
Data Layer	PostgreSQL; users, trainers, trainees, trainings, enrollment/request, learning, attendance, certificate and related records

Request flow: Browser → React application → Express API → authorization/business rules → PostgreSQL → API response → React UI.

For enrollment, the backend must perform the final eligibility and authorization checks even when the frontend already filters the visible training list. This prevents a user from bypassing department restrictions by manually calling an API endpoint.

10. System Components

10.1 Frontend

• React.js application providing public pages and authenticated role-specific dashboards.

• React Router DOM for route navigation and protected/role-aware routes.

• Tailwind CSS and reusable layout/components for the professional MinT interface.

• Axios or equivalent HTTP service layer for API communication.

• Toast notifications and confirmation dialogs for user actions.

• Language support for English and Amharic.

• Responsive layouts for desktop and smaller screens.

10.2 Backend API

• Node.js runtime with Express.js web framework.

• Authentication routes for registration/login and protected API routes.

• Middleware for JWT verification and role-based authorization.

• Controllers/services for training, trainer, trainee, enrollment, learning, attendance, certificate and report operations.

• Server-side validation of department eligibility and enrollment status.

• Database access through the project's PostgreSQL data-access layer.

10.3 PostgreSQL Database

PostgreSQL is used as the persistent relational database. The database stores application users and role-specific records together with training programs, department assignments, enrollment requests, learning records, attendance, and certificates. Database-backed records are used by the administrative request-management interface rather than hard-coded sample data.

11. Functional Requirements

FR-01 — The system shall allow registered users to authenticate using valid credentials.

FR-02 — The system shall issue and validate authenticated JWT sessions for protected API operations.

FR-03 — The system shall restrict administrative, trainer, and trainee functions according to role permissions.

FR-04 — The system shall allow administrators to create, update, view, and manage training programs.

FR-05 — The system shall allow administrators to create trainer accounts without pre-populating another user's email or password.

FR-06 — The system shall allow trainees to register with department information.

FR-07 — The system shall support training assignments to one department, multiple departments, or all departments.

FR-08 — The system shall determine trainee eligibility using the training's actual department assignment.

FR-09 — The system shall allow eligible trainees to submit enrollment requests.

FR-10 — A newly submitted enrollment request shall have Pending status.

FR-11 — Administrators shall be able to approve or reject Pending requests.

FR-12 — Trainees shall be able to cancel their own Pending requests.

FR-13 — A trainee shall not be able to cancel another trainee's request.

FR-14 — The normal Pending cancellation function shall not remove an Approved enrollment.

FR-15 — Administrators shall be able to view trainee, department, training, request date, status, and relevant action information.

FR-16 — Trainers and authorized administrators shall be able to create and publish permitted learning materials.

FR-17 — Trainees shall only see published learning content for trainings they are permitted to access.

FR-18 — The system shall record learning progress and support the configured completion threshold.

FR-19 — The system shall support attendance management for training participants.

FR-20 — The system shall generate/store certificates for trainees who satisfy the configured completion requirements.

FR-21 — The system shall provide reports and dashboard information based on database records.

FR-22 — The public website shall provide a professional contact interface and shall not falsely claim that an unconfigured message was delivered.

FR-23 — The public website shall provide working configured social-media links.

FR-24 — The interface shall support English and Amharic user-facing text.

12. Training Assignment and Enrollment Workflow

12.1 Department Assignment Rules

Assignment	Expected trainee access
One department	Only trainees belonging to that department can view/request the training.
Multiple departments	Trainees belonging to any assigned department can view/request the training.
All departments	Trainees from every supported department can view/request the training.

The implementation must not solve an eligibility error by simply removing department restrictions. The frontend filtering, API response, backend validation, database representation, and enrollment-request logic must agree.

12.2 Enrollment Status Workflow

Trainee requests training → Pending → Admin approves → Approved
or Trainee requests training → Pending → Admin rejects → Rejected
or Trainee requests training → Pending → Trainee cancels → Cancelled/Removed according to the database design.

Administrators must have visibility into the complete request workflow. The request-management page should use actual database data and provide appropriate search/filter functionality.

13. Learning Content and Progress

• Trainers and authorized administrators publish modules and lessons for permitted training programs.

• Trainees access only content that is marked as published and belongs to an approved/eligible training.

• The trainee can read/view learning material and use a Next action to progress through the learning sequence.

• The system records progress and presents it as a percentage.

• The configured completion threshold is 75% for unlocking the required questions/completion stage, based on the current project workflow.

• After the required completion conditions are satisfied, the trainee becomes eligible for certificate generation.

14. Attendance and Certificates

Attendance is included as a training-management function so that administrators/trainers can maintain participation records alongside enrollment and learning records. Attendance data can support reporting and provide an additional record of participation.

Certificates are generated from database-backed trainee and training information. A certificate record should identify the trainee, completed training, issue information, and a unique certificate number. Sensitive information such as passwords must never appear in certificate or administrative views.

15. Reports and Dashboard Requirements

• Dashboard statistics should be derived from current database records.

• Administrators should be able to monitor trainers, trainees, trainings, enrollment requests, attendance, and certificates according to existing permissions.

• Enrollment monitoring should include trainee name, email, department, training, request date, status, and available approval/rejection/action information.

• Reports should avoid exposing unnecessary sensitive information.

• Role-specific dashboards should remain visually consistent and responsive.

16. Non-Functional Requirements

16.1 Usability

• The interface should be professional, clean, readable, and consistent with MinT government-style presentation.

• Navigation, buttons, forms, notifications, tables, and dashboards should use consistent typography and spacing.

• The public pages should use the public header/footer, while authenticated dashboards use their appropriate role layout.

• The interface should provide clear feedback after important actions.

16.2 Performance

Normal API operations should respond within a practical time under normal local/application load. Database queries and frontend rendering should avoid unnecessary repeated requests and expensive operations.

16.3 Reliability

The system should preserve consistent enrollment and training records. Important status transitions must be validated on the server and reflected correctly in the database and user interface.

16.4 Maintainability

Existing working components, services, routes, controllers, layouts, and database structures should be reused where appropriate. Changes should address root causes rather than adding UI-only workarounds.

16.5 Responsiveness

Public pages and dashboards should remain usable across common desktop, tablet, and smaller screen widths. Sidebar navigation and active/hover states should remain visually clear without distracting effects.

17. Security Requirements

• Users must authenticate before accessing protected resources.

• JWT tokens must be validated for protected API requests.

• Role-based authorization must be enforced on the backend, not only in the frontend.

• Passwords must be stored securely using bcrypt or the existing secure password-hashing implementation.

• Passwords and other credentials must never be exposed in the UI, reports, API responses, or source code.

• A trainee may access and modify only their own permitted request records.

• A trainee must never approve their own request or another trainee's request.

• Administrative functions must be limited to authorized administrators.

• User-provided data must be validated before database operations.

• Department eligibility must be checked server-side to prevent bypassing frontend restrictions.

18. Error Handling

• Invalid authentication credentials.

• Missing or invalid JWT token.

• Unauthorized role access.

• Invalid or incomplete training data.

• Training not assigned to the trainee's department.

• Invalid enrollment status transition.

• Attempt to cancel another trainee's request.

• Attempt to cancel an Approved request through the Pending cancellation function.

• Database connection or query errors.

• Unexpected server errors.

• Unconfigured contact-message delivery.

Error messages should explain the user-facing problem without exposing stack traces, passwords, database credentials, or other internal sensitive details.

19. Testing Requirements

19.1 Unit Testing

• Authentication and authorization helpers.

• Department eligibility logic.

• Enrollment status transitions.

• Progress calculation.

• Certificate eligibility logic.

19.2 Integration Testing

• React frontend to Express API communication.

• Express API to PostgreSQL operations.

• Admin approval/rejection actions.

• Trainee cancellation and database status updates.

19.3 System Testing

Scenario	Expected result
Single department	Computer Science trainee can request; an IT trainee cannot.
Multiple departments	Trainees from each assigned department can request; others cannot.
All departments	Eligible trainees from every supported department can request; no false department error.
Pending request	Trainee can view and cancel their own request before approval.
Approved request	Trainee cannot remove it through the Pending cancellation function.
Admin monitoring	Admin sees trainee, department, training, request date and current status.

Trainer creation	New trainer form starts empty and does not expose existing credentials.
Public Training Programs	Public header/footer is shown instead of a dashboard sidebar.
Language	English and Amharic interface text can be switched consistently.

19.4 Final Verification

• Check browser console for frontend errors.

• Check backend/server logs for runtime errors.

• Inspect API responses for correct status codes and data.

• Verify database updates after enrollment, approval, rejection, and cancellation.

• Test Admin, Trainer, and Trainee separately.

• Confirm that existing authentication, learning, attendance, certificates, reports, and dashboard functionality still works.

20. Future Enhancements

• Expanded Amharic translation coverage and language administration.

• Email or notification support for enrollment approval/rejection and training announcements.

• More advanced attendance analytics and participant reports.

• Training feedback and evaluation forms.

• Improved certificate verification and public verification workflow.

• Automated reminders for pending requests and unfinished learning.

• Additional learning-content formats and richer media support.

• More detailed audit logs for administrative actions.

• Mobile-focused interface improvements.

• Integration with other authorized MinT information systems where appropriate.

21. Conclusion

The MinT Training Management System provides a centralized approach to managing short-term training programs for the Ministry of Innovation and Technology. Its role-based structure separates administrative, trainer, and trainee responsibilities while maintaining a common data foundation in PostgreSQL.

The department-aware assignment model, controlled enrollment workflow, learning-progress process, attendance, certificates, and reporting functions address the major operational requirements of the project. The system is designed to remain maintainable by reusing existing components and enforcing important business rules in the backend.

The documentation structure follows the supplied example's organization—introduction, background, problem, purpose, objectives, scope, users, architecture, components, functional requirements, non-functional requirements, security, error handling, testing, and future enhancements—while adapting the content to the MinT Training Management System.

Reference basis: This document was prepared using the supplied documentation format as the structural reference and the current MinT system requirements/project context for the system-specific content.